

COURSE TITLE : DESIGN & STRUCTURE OF FABRICS
COURSE CODE : 3061
COURSE CATEGORY : B
PERIODS/WEEK : 5
PERIODS/SEMESTER : 75
CREDITS : 5

TIME SCHEDULE

Module	Topics	Periods
1	Plain & Twill Designs	23
2	Towelling Fabrics	19
3	Compound Structures	18
4	Crepe and Perforated Fabrics	15
TOTAL		75

COURSE OBJECTIVE

Module	GO	Student will be able to
1	1	Understand Textile designs, draft, lifting plan & tie up and treadling plan.
	2	Know plain design & its modifications.
	3	Know Twill design & its classifications & modifications
2	1	Understand huck-a-back designs
	2	Understand honey comb designs
	3	Know about different pile designs
3	1	Know about bedford cord & pique designs
	2	Understand double cloth principle designs
	3	Know extra wrap and extra weft figuring designs.
4	1	Understand different types of crepe fabrics & crepe designs
	2	Understand Mock leno designs.
	3	Know cross weaving, Gauze & leno designs.

SPECIFIC OUTCOME

MODULE I PLAIN & TWILL DESIGNS

1.1.0 Understand Textile designs, draft, lifting plan

- 1.1.1 Identify the use of point paper for textile designing.
- 1.1.2 Explain how the way in which designs, draft, lifting plan, denting plan etc are produced in design paper.
- 1.1.3 Classify different drafts like straight , broken , pointed and mixed drafts

1.2.0 Know plain design & its modifications.

- 1.2.1 Define plain design & its decorative & structural modifications
- 1.2.2 Illustrate Wrap Rib designs
- 1.2.3 Illustrate Weft Rib designs
- 1.2.4 Illustrate mat designs

1.3.0 Know Twill design , its classifications & modifications

- 1.3.1 State general features of twill designs.
- 1.3.2 Draw twill designs using different motifs.
- 1.3.3 Illustrate the general classification of twill designs
- 1.3.4 Identify the modifications of twill designs
- 1.3.5 Design the drafts applicable for the above designs
- 1.3.6 State the influence of twist of yarn on the prominence of twill lines.

MODULE II TOWELLING FABRICS

2.1.0 Understand huck-a-back designs

- 2.1.1 State the general requirements of a toweling fabric.
- 2.1.2 Describe Huck – a – back designs
- 2.1.3 Design Devon's Huck-a -back, standard Huck-a- back and Honey comb Huck- a –back designs.

2.2.0 Understand honey comb designs

- 2.2.1 Illustrate Honey comb designs
- 2.2.2 Design ordinary and stitched honey comb designs
- 2.2.3 Describe Brighton's Honey comb designs & distinguish them with ordinary honey comb.

2.3.0 Know about different pile designs

- 2.2.1 State the classification of pile designs & explain how the piles are produced in different pile fabrics
- 2.2.2 Differentiate warp pile and weft pile structures.
- 2.2.3 Design 3 pick,4 pick and 5 pick terry pile designs.
- 2.2.4 Identify drafts specially applicable to the above designs.
- 2.2.5 State the general features of pile fabrics
- 2.2.6 Design 2 pick to a wire, 3 pick to a wire and fast warp pile Designs.
- 2.2.7 Draw velvet, velveteen designs

MODULE III COMPOUND STRUCTURED DESIGNS

3.1.0 Know about bedford cord & pique designs

- 3.1.1 State the general features of Bedford cord fabrics.
- 3.1.2 Design plain face Bedford cord and twill face Bedford cord.
- 3.1.3 Draw wadded bed ford cords and develop their drafts .
- 3.1.4 State the functions of cutting ends, face ends and wadding ends
- 3.1.5 State the general features of pique fabrics.
- 3.1.6 Differentiate pique fabrics from Bedford cord fabrics.
- 3.1.6 State the functions of face, cutting, wadding and back picks.
- 3.1.7 Design coarse cut , fine cut , wadded & backed pique designs.

3.2.0 Understand double cloth principle designs

- 3.2.1 State the general features of double cloth designs.
- 3.2.2 Describe and construct double cloth, double width cloth, tubular cloth and interchanging double cloth.

3.3.0 Know extra wrap and extra weft figuring designs.

- 3.3.1 Illustrate extra wrap and extra weft figuring for ornamentation of fabric
- 3.3.2 Describe stripe figuring, all over figuring and spot figuring .
- 3.3.3 List the feature of toilet quilt fabrics.
- 3.3.4 Draw examples for 2 pick and 3 pick toilet quilts.

MODULE IV CREPE & PERFORATED FABRICS

4.1.0 Understand different types of crepe fabrics & crepe designs

- 4.1.1 State general characteristics of crepe fabrics.
- 4.1.2 List different methods used to impart crepe effect on fabric
- 4.1.3 Illustrate crepe designs.
- 4.1.4 Identify different methods of construction of crepe designs.
- 4.1.5 State the principle of weaving seer-sucker

4.2.0 Understand Mock leno designs.

- 4.2.1 Define perforated fabrics and their use.
- 4.2.2 Explain how to produce 3 X 3, 4 X 4, 5 X 5 mock leno designs.
- 4.2.3 Identify the draft and denting plan for these designs.

4.3.0 Know cross weaving, Gauze & leno designs.

- 4.3.1 State the principle of cross weaving.
- 4.3.2 Describe the formation of cross sheds, and open shed on bottom doup healds.
- 4.3.3 Draw the diagram of the cloth structure, drafting and lifting plan for plain gauze
- 4.3.4 Draw the diagram of the cloth structure, drafting and lifting plan for simple leno.
- 4.3.5 Apply designs for representing the above designs.
- 4.3.6 Identify the sheds in leno weaving

CONTENT DETAILS

MODULE I PLAIN AND TWILL DESIGNS

Application of Point paper in textile Designing - count of point paper and definition of design, draft, lifting plan and denting plan. - Methods of representing design, draft, lifting plan, Tie-up & treadling plan and denting plan.- Types of draft - straight draft, broken draft, pointed draft, mixed draft. Structure of plain design- its construction- modification of plain design –structural & decorative - warp rib designs, weft rib designs and mat designs - rib and cord effects,- Twill designs-construction & motif - General features of twill designs - Classification - Simple twill- Compound twill - Left hand twill- Right hand twill - Warp faced twill.- Weft faced twill- Even faced twill- Steep twill- Flat twill. Modification of twill designs - Satin and sateen designs- Warp Cork- screw and weft Cork-screw designs- Combined twills- Figured twill- horizontal & vertical Zig zag twills- Diamond twills- Broken twills- Broken and reversed twill- Drafts for the above designs- Influence of yarn twist on the prominence of twill lines.

MODULE II TOWELLING FABRICS

General requirements of toweling fabrics - designs used for toweling fabrics- Features of Huck – a – Back fabrics- Construction of Standard Huck- a - Back (10 X 10) - Devon's Huck– a– Back - Honey comb Huck – a- Back designs - Reversible Huck – a- back design - Drafts applicable to the above designs. Construction of ordinary honey comb – stitched honey comb - Brighton's honey comb designs - Drafts applicable to the above designs- Features of pile fabric- Classification of pile designs - Construction of 3-pick terry, 4-pick terry and, 5-pick terry pile designs. - - Drafts applicable - - True warp pile - Two pick to a wire - Three pick to a wire, Fast true warp pile, - Construction of Velvet- Drafts applicable - construction of Weft pile designs - Construct Plain back, Twill back pile designs, Corded velveteen designs- Differentiate warp pile and weft pile structures, velvet & velveteen -Importance of cutting and finishing in the production of cut weft piles.

MODULE III COMPOUND STRUCTURES

Differentiate between simple and compound structures - General features of Bedford cord fabrics- Construction of plain face Bedford cord, Twill face Bedford cord, . Wadded Bedford cord and develop their drafts- Functions of cutting ends, face ends and wadding ends- Features of Pique fabrics- - Construction of various types of pique fabrics - Coarse cut pique, Fine cut pique- Wadded pique, Backed pique- Functions of cutting, face, wadding and back picks - System of Drafting and Denting applicable to above designs- Additional attachments required in a loom to weave Bedford cord and Pique fabrics- Importance of Double cloth weaving - Construct Double cloth, Double width cloth, Tubular - Interchanging Double cloth -- Drafts applicable - Construct extra warp and extra weft designs - Importance of extra warp and extra weft figuring in ornamentation of fabrics- All over figuring, Stripe figuring and Spot figuring- Features of Toilet Quilt fabrics- Construct 2 pick, and 3 pick Toilet Quilt designs.

MODULE IV CREPE & PERFORATED FABRICS

Features of crepe fabrics- Different methods of constructing crepe fabrics - Different methods of constructing crepe designs- By adding marks to satin designs, Combining plain threads with a floating design, Inserting one designs over another, Reversing a small repeat - Principle of seer sucker fabrics. Define perforated fabrics - Application of Perforated fabrics - Construct 3 X 3, 4 X 4, 5 X 5 Mock- Leno designs - Draft and Denting plan- Speciality of denting for Mock-leno designs- Principles of Cross weaving- Application of Doup Healds-types of douphealds- Various types of sheds formed in cross weaving- Plain shed, Open shed and Cross shed - Differentiate Gauze and Leno - Construction of plain gauze. - ordinary leno- thread diagram & graphical representation - Drafting and liftig plan for Plain Gauze, & leno.

REFERENCE BOOKS

1. NISBERT - Grammar of Textile Design Mahajan Publishers, Ahammedabad
2. WATSON - Textile Design and Color Do
3. WATSON - Advanced Textile Design - Do.
4. TEXTILE INSTITUTE. Cloth Construction -
5. Gokarneshan, N. Fabric Structure and Design